

FINAL YEAR ENGINEERING PROJECT REVIEW

Campus Library Management System: A Dual-Client Native & Web Architecture

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1. Project Abstract & Executive Overview

The Campus Library Management System is a production-grade enterprise application designed to resolve the inefficiencies of traditional manual and legacy desktop library management. The solution bridges student convenience and administrative security by uniting two dedicated client platforms through a robust RESTful API backend:

CORE STACK

Node.js + Kotlin

Zero extra frameworks

DATABASE

PostgreSQL

Native 5432 & PGlite

REAL DATA

Open Library

Internet Archive API

BULK IMPORT

MS Excel (.xlsx)

Multi-student batch

2. Architectural Constraints & Stack Adherence

Per strict academic project specifications, this project has been built strictly without third-party frameworks like React, Angular, Vue, Spring, or Django. It adheres strictly to the sanctioned stack:

Component / Layer	Sanctioned Technology	Implementation Details & Engineering Scope
Student Mobile Client	Native Android / Kotlin	Retrofit2, OkHttp3, Material 3, ViewBinding, SharedPreferences JWT Session.
Admin Web Portal	Vanilla HTML5 / CSS3 / JS	Zero frontend framework, CSS Grid/Flexbox, dynamic SPA routing, modal workflows.
API Backend	Node.js (v24) + Express	RESTful API architecture, JWT token authentication, Multer file upload parsing.
Relational Database	PostgreSQL Dual-Engine	ACID-compliant tables, foreign keys, ON DELETE CASCADE, PGlite persistent fallback.
External Data Integration	Open Library (Internet Archive)	Direct REST integration with openlibrary.org for live book search, covers, & ISBN auto-fill
Bulk Spreadsheet Engine	SheetJS (xlsx) Engine	Pure binary parser and workbook generator for Microsoft Excel (.xlsx/.xls/.csv).

3. System Architecture & High-Level Topology

The system follows a classic 3-Tier Enterprise Client-Server Architecture where both clients interact with a unified REST API backend, enforcing centralized security, business logic, and transactional database integrity.

STUDENT MOBILE CLIENT (KOTLIN)

- Material 3 UI with Live Search & Category Chips
- My Loans dashboard with Due Date & Overdue countdowns
- Strict View-Only Access (Cannot issue or tamper with loans)

LIBRARIAN ADMIN PORTAL (VANILLA WEB)

- Circulation Desk: Issue books & record returns with fines
- Open Library Explorer: Live search & 1-click import
- Excel Student Batch: Download template & upload roster

UNIFIED REST API BACKEND & DATABASE TIER (NODE.JS + POSTGRESQL)

- Middleware: authenticateToken (JWT verification) | requireLibrarian (403 Forbidden enforcement)
- PostgreSQL Relational Tables: users (students & librarians), books (catalog & stock), borrow_records (active & history)
- External Gateway: openlibrary.org API client with error handling & cover image parsing

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Section II: Relational Schema, Security Model & Novel Features

CORE FOCUS: Database Normalization, RBAC Protection & External API Integration

4. Relational Database Design & Schema Specifications

The database is engineered in third normal form (3NF) to eliminate data redundancy and preserve referential integrity across all transactional operations. Dual-engine support enables native PostgreSQL or embedded zero-config PGlite.

Table: users

Stores student and librarian accounts. Passwords securely hashed via bcryptjs (10 salt rounds).

`id (PK SERIAL), name (VARCHAR 100), email (VARCHAR 150 UNIQUE), student_id (VARCHAR 50 UNIQUE), password_hash (VARCHAR 255), role (VARCHAR: student | librarian), department (VARCHAR 100), phone (VARCHAR 20), created_at (TIMESTAMP)`

Table: books

Inventory catalog holding bibliographic details, total/available copies, shelf tags, and Open Library artwork.

`id (PK SERIAL), isbn (VARCHAR 50 UNIQUE), title (VARCHAR 255), author (VARCHAR 255), genre (VARCHAR 100), published_year (INT), total_copies (INT DEFAULT 1), available_copies (INT DEFAULT 1), shelf_location (VARCHAR 50), description (TEXT), cover_url (TEXT), created_at (TIMESTAMP)`

Table: borrow_records

Maintains lending ledger with foreign keys to users and books. Enforces loan periods, return dates, and fine amounts.

`id (PK SERIAL), user_id (FK -> users.id ON DELETE CASCADE), book_id (FK -> books.id ON DELETE RESTRICT), issue_date (DATE), due_date (DATE), return_date (DATE NULL), status (VARCHAR: active | returned | overdue), fine_amount (NUMERIC(8,2) DEFAULT 0.00)`

5. Role-Based Access Control (RBAC) & Security Enforcement

A core academic requirement is that student loan issuance is strictly restricted to librarians. The system enforces multi-tier security:

1. Stateless JWT Authentication

Users receive a signed JSON Web Token upon validating credentials at `/api/auth/login`. The payload encapsulates user ID, email, student ID, and role. Token expiration is configured to 7 days.

2. requireLibrarian Middleware (Strict 403 Forbidden)

Protected administrative routes (POST `/api/borrow/issue`, POST `/api/borrow/return/:id`, POST `/api/books`, POST `/api/books/openlibrary/import`, POST `/api/auth/students/upload`) intercept requests. If `req.user.role !== "librarian"`, the request is immediately terminated with HTTP 403 Forbidden.

3. Data Isolation for Students

Students can only query their own loan records via GET `/api/borrow/my-books`, which binds exclusively to `req.user.id` extracted from the cryptographically verified JWT token, preventing cross-student data leakage.

4. Cryptographic Password Hashing

Passwords are never stored in plain text. All passwords (registered manually or imported via Excel) undergo salted one-way hashing with bcryptjs using 10 rounds of salt before database insertion.

6. Novel Feature Implementations

A. Open Library Integration by Internet Archive (openlibrary.org)

Instead of static, mock catalog entries, the application features an enterprise integration with the Internet Archive Open Library API:

- Live Catalog Search: Librarians query millions of real published books directly inside the portal (title, author, year, edition, and cover thumbnail).
- 1-Click Catalog Import: Automatically imports selected titles into PostgreSQL with custom copies and shelf location.
- ISBN Auto-Fill: When adding individual titles, entering a 10 or 13-digit ISBN automatically retrieves full bibliographic metadata.

B. Bulk Student Onboarding via Microsoft Excel (.xlsx / .csv)

To support real-world university rollouts with hundreds of incoming students, the system provides bulk onboarding:

- Genuine Binary Excel Template: Downloadable via GET `/api/auth/students/template` as a real Microsoft Excel .xlsx workbook.
- Smart Column Mapping & Validation: Handles flexible headers (Full Name, Email, Student ID, Phone, Department, Initial Password).
- Duplicate Prevention & Batch Report: Skips duplicate student IDs/emails and provides an exact summary report (created vs skipped).

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Section III: Verification Matrix, Test Results & Academic Sign-Off

STATUS: Complete Integration Verified | NATIVE MOBILE + WEB + API VALIDATED

7. Comprehensive Test & Verification Matrix

The complete application has undergone exhaustive end-to-end integration and security testing. All test scenarios have passed successfully:

ID	Test Scenario	Execution & Payload	Expected & Verified Outcome	Status
TC-01	Librarian Authentication	POST /api/auth/login (admin@library.com)	HTTP 200 OK, JWT issued with role: librarian	PASSED
TC-02	Demo Student Login	POST /api/auth/login (student@college.edu)	HTTP 200 OK, JWT issued with role: student	PASSED
TC-03	Student Loan Tamper Block	Student token calling POST /api/borrow/issue	HTTP 403 Forbidden ("Librarian access required")	PASSED
TC-04	Student Catalog Retrieval	GET /api/books with student JWT	HTTP 200 OK, 8 real Open Library books loaded	PASSED
TC-05	Binary Excel Template Download	GET /api/auth/students/template	HTTP 200 OK, binary application/vnd.openxmlformats(.xlsx)	PASSED
TC-06	Open Library Live Proxy	GET /api/books/openlibrary/search?n=borrow+notter	HTTP 200 OK, live data & cover URLs from openlibrary.org	PASSED
TC-07	Excel Batch Student Import	POST /api/auth/students/upload with .xlsx file	HTTP 201 Created, batch created/skipped summary returned	PASSED
TC-08	Circulation Fine Calculation	POST /api/borrow/return/:id on overdue book	HTTP 200 OK, fine computed (\$0.50/day overdue), stock +1	PASSED
TC-09	Native Android Mobile Execution	Kotlin app running on Android Emulator	Full catalog rendering with shelf location & live stock	PASSED

8. Current Deployed State & Demonstration Credentials

The live test environment is operational and configured for evaluation:

CHIEF LIBRARIAN (ADMIN) admin@library.com Password: admin123 (LIB-001) Admin Portal (http://localhost:5000)	DEMO STUDENT student@college.edu Password: student123 (STU-1001) Native Android App & Student APIs	LIVE CATALOG 8 Real Engineering Books with Open Library Cover Art Sourced from Open Library (Internet Archive)
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9. Conclusion & Project Sign-Off

The Campus Library Management System meets all engineering, functional, and non-functional requirements specified for the Final Year Project. The solution provides a modular, highly scalable foundation that strictly honors the assigned technology stack and delivers tangible real-world utility through Open Library live cataloging and bulk spreadsheet onboarding.

Candidate / Project Lead

Final Year B.Tech / B.E.

Project Internal Guide

Dept. of Computer Science & Engg.

Head of Department / Reviewer

Academic Review Committee